THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE UNIT TEST SET B

Chapters 9–13: **Cell: The Building Block of Life, Tissues in Action, Patterns in Life: Diversity and Classification, Reproduction: How Life Continues, Earth as a System: Energy, Matter and Life** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. The plastids that give colour to flowers and fruits are: 1

(a) chromoplasts **(b)** chloroplasts **(c)** lysosomes **(d)** leucoplasts

2. The movement of water across a selectively permeable membrane is called: 1

(a) photosynthesis **(b)** osmosis
(c) diffusion of gases **(d)** transpiration

3. Blood is classified as a: 1

(a) nervous tissue **(b)** epithelial tissue
(c) connective tissue **(d)** muscular tissue

4. A group of similar cells performing a specific function is called a: 1

(a) organ **(b)** tissue **(c)** system **(d)** organelle

5. Meristematic cells are characterised by: 1

(a) thick cell walls **(b)** large vacuoles
(c) wide intercellular spaces **(d)** dense cytoplasm and a prominent nucleus

6. The variety of living organisms on the Earth is called: 1

(a) evolution **(b)** ecosystem **(c)** biodiversity **(d)** classification

7. The group of vertebrates that is cold-blooded and breathes through gills is: 1

(a) fishes **(b)** birds **(c)** mammals **(d)** reptiles

8. Spirogyra is an example of: 1

(a) pteridophyta **(b)** angiosperm
(c) bryophyta **(d)** algae (thallophyta)

9. Bright coloured, fragrant flowers with nectar are generally pollinated by: 1

(a) water **(b)** wind **(c)** insects **(d)** none

10. Pollen grains are produced in the: 1

(a) anther **(b)** style **(c)** ovary **(d)** stigma

11. Greenhouse gases include: 1

(a) carbon dioxide, methane and water vapour **(b)** neon and hydrogen
(c) nitrogen and oxygen **(d)** argon and helium

12. Conversion of water vapour into droplets is: 1

(a) infiltration **(b)** transpiration **(c)** evaporation **(d)** condensation

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 10–15 words each.

13. What is the function of the cell membrane? 2

14. What is differentiation? 2

15. Why are cyanobacteria important? 2

16. Distinguish between asexual and sexual reproduction (one point each). 2

17. Name two greenhouse gases and a human source for each. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 30–40 words each.

18. Compare prokaryotic and eukaryotic cells in three points. 3

19. Describe the structure of xylem. 3

20. Give three examples of seed dispersal methods and plants that use them. 3

21. Calculate the energy received by a 5 m^2 solar panel in 2 hours at 1 kW m^{-2} insolation. 3

22. Explain how the Earth's tilt and shape cause seasons and uneven heating. 3

OR

Why is Venus hotter than Mercury although Mercury is nearer to the Sun? 3

23. Explain how mitochondria and plastids support the idea that they evolved from bacteria. 3

OR

State three differences between plant and animal cells. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

24. (i) Draw and label a plant cell. 5
(ii) State two features that distinguish it from an animal cell.
(iii) Name the organelle involved in photosynthesis.

OR

Case study: A student is to make a table of the organelles: mitochondria, ER, Golgi, lysosomes, ribosomes. 5
(i) Give the function of each.
(ii) Which two have their own DNA?
(iii) Which makes proteins?
(iv) Which digests waste?

25. (i) What is biodiversity? 5
(ii) Why should it be conserved?
(iii) Give two examples of interdependence among organisms.

OR

(i) Classify the plant kingdom with one example each. 5
(ii) Give one feature of each division.

END OF QUESTION PAPER — Total: 50 Marks